

Smart Library Request Workflow in ServiceNow

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1.INTRODUCTION:

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OVERVIEW OF THE PROJECT:

- The Smart Library Request Workflow in ServiceNow project focuses on improving the efficiency of library management in educational institutions.
- In many libraries, book requests and approvals are handled manually, which can lead to delays and errors. Students often find it difficult to know the availability of books, and librarians face challenges in tracking borrow requests.
- This project provides a digital solution where students can request books through the system. Librarians can review and approve or reject the requests using an automated workflow.
- This system improves transparency, reduces manual effort, and makes library management more efficient.

2. IDENTIFICATION PHASE:

INTRODUCTION:

- Many educational institutions still manage library activities manually. Students need to check book availability by directly visiting the library, and librarians must manually track borrow and return records.
- This manual process can lead to delays, lack of transparency, and difficulty in maintaining accurate records.

PROBLEM IDENTIFICATION:

- Students often face problems in knowing which books are available. Librarians also find it difficult to track borrow requests, approve them, and update the book status manually.
- This process consumes time and may result in errors.

OBJECTIVES OF THE PROJECT:

- To create a digital library request system.
- To allow students to request books online.
- To enable librarians to approve or reject requests.
- To automate the process of updating book status.

3. REQUIREMENTS ANALYSIS:

- Requirement analysis is an important stage where the needs and requirements of the system are identified before development begins.

Functional Requirements:

- Students should be able to request books through the system.
- Librarians should be able to manage book records.
- Librarians should approve or reject borrow requests.
- Book status should update automatically when issued or returned.
- Reports should show the most borrowed books.

Non-Functional Requirements:

- The system should be user friendly.
- Data should be secure and reliable.
- The system should provide quick response time.

Software Requirements:

- ServiceNow platform
- Google Sheets or Microsoft Excel
- Web browser such as Chrome or Edge

Hardware Requirements:

- Computer or laptop
- Internet connection
- ServiceNow instance access

4. PROJECT PLANNING PHASE:

- The Project planning defines how the project will be executed.

Project Schedule:

- The project is divided into several stages such as:
 - ❖ Problem identification
 - ❖ Requirement analysis
 - ❖ System design
 - ❖ Development and implementation
 - ❖ Testing
 - ❖ Documentation
 - ❖ Final demonstration

Planning Activities:

- Creating roles for students and librarians
- Designing tables for storing book and request data
- Creating workflows for approval process
- Configuring security rules
- Creating reports and testing the system

Risk Management:

- Some possible risks during project implementation include:
 - Incorrect field mapping during transform map creation
 - Data format errors in the import file
 - Configuration mistakes in ServiceNow tables

Resources Required:

- ServiceNow Personal Developer Instance
- Sample library data
- Documentation tools

5. PROJECT DEVELOPMENT PHASE:

- ✓ The development phase focuses on implementing the actual solution in the ServiceNow platform.

Step 1: Creating Roles

- Two roles are created:
 - ✓ Student role – used for requesting books
 - ✓ Librarian role – used for managing books and approvals

Step 2: Creating Tables

- Two custom tables are created:
 - ✓ Book Table – stores book details
 - ✓ Borrow Request Table – stores book request information

Step 3: Adding Fields

Fields are added to the tables.

- Book Table Fields
 - ✓ Title
 - ✓ Author

- ✓ ISBN
- ✓ Status

- Borrow Request Table Fields:
 - ✓ Requested
 - ✓ Book
 - ✓ Request Date
 - ✓ Status

Step 4: Configuring Reference Qualifier

- A reference qualifier is added so that students can only select books that are available.

Step 5: Implementing Workflow

- Using Flow Designer, a workflow is created where:
 - Students submit borrow requests
 - Librarians approve or reject requests
 - If approved, the book status is automatically changed to Issued

6. PROJECT DESIGN PHASE:

- This phase describes the system structure and how components interact

System Design Overview:

- The system is designed to automate library operations using ServiceNow features.

Key Design Components:

✓ Roles

Roles define the permissions for students and librarians.

✓ Tables

Tables store book information and borrow request records.

✓ Flow Designer

Used to automate the approval process and update book status

✓ UI Policies

Used to control form behavior and improve usability.

✓ **Access Control Rules**

Used to provide security and restrict access based on roles.

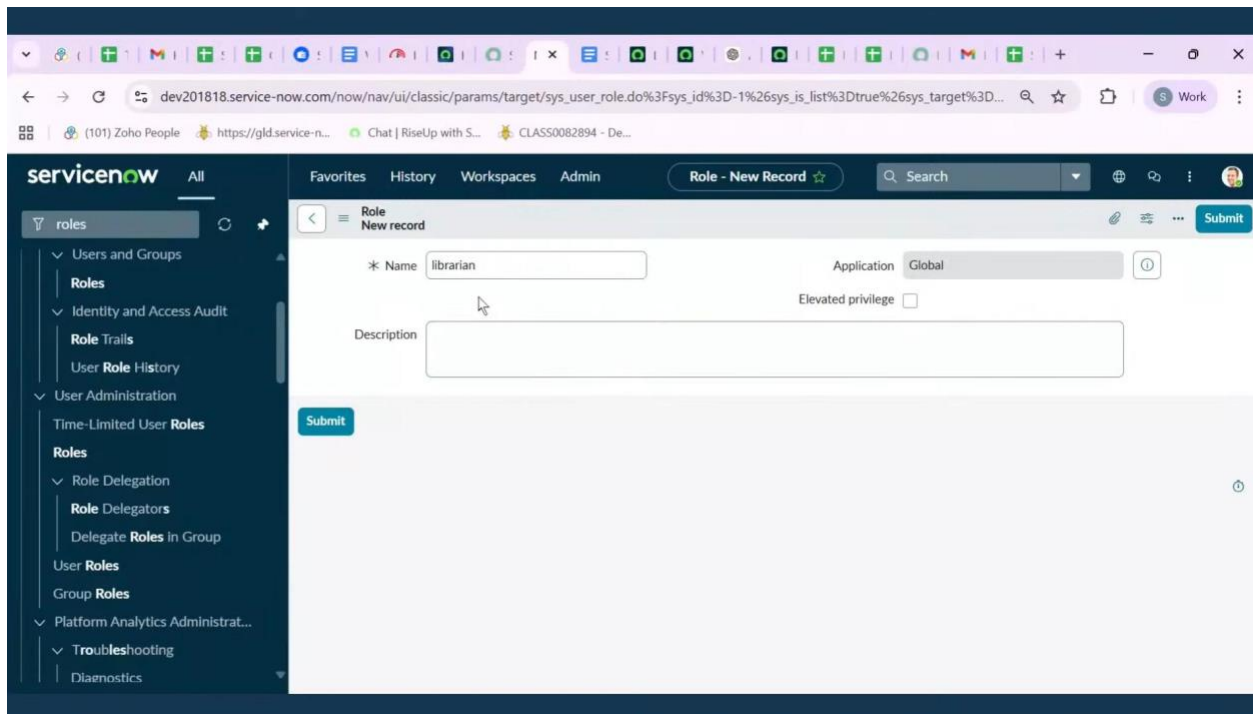
✓ **System Workflow**

- ❖ Student submits borrow request.
- ❖ Request is sent to librarian.
- ❖ Librarian approves or rejects the request.
- ❖ Book status changes automatically.
- ❖ Reports show book usage statistics.

7. PROJECT DOCUMENTATION:

Project documentation is an essential part of the project as it records all the information related to the project development process

Step 1: Create Roles



The screenshot shows the ServiceNow 'Role - New Record' form. The left sidebar contains a navigation menu with the following items: roles, Users and Groups, Roles, Identity and Access Audit, Role Trails, User Role History, User Administration, Time-Limited User Roles, Role Delegation, Role Delegators, Delegate Roles in Group, User Roles, Group Roles, Platform Analytics Administrat..., Troubleshooting, and Diagnostics. The main form area has a header with 'Role - New Record' and a search bar. Below the header, there are fields for 'Name' (with a required asterisk) containing 'librarian', 'Application' set to 'Global', and 'Elevated privilege' which is an unchecked checkbox. There is also a 'Description' field. A 'Submit' button is located at the bottom left of the form area.

dev201818.service-now.com/now/nav/ui/classic/params/target/sys_user_role.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3D...

roles

- Users and Groups
 - Roles**
- Identity and Access Audit
 - Role Trails
 - User Role History
- User Administration
 - Time-Limited User Roles
 - Roles**
 - Role Delegation
 - Role Delegators
 - Delegate Roles In Group
 - User Roles
 - Group Roles
- Platform Analytics Administrat...
 - Troubleshooting
 - Diagnostics

Role - New Record

Name: student

Application: Global

Elevated privilege: ☐

Description:

Submit

Step 2: Add Fields

dev201818.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3D...

tables

- Log Protection
 - Protected Table Log
- System Archiving
 - Archive Tables**
 - Archive Audit Result
 - Archive Knowledge Use
- System Clone
 - Clone Definition
 - Exclude Tables
- System Definition
 - Tables**
 - Tables & Columns
 - Table Rotations
 - Decision Tables
 - Remote Tables
 - Tables

Table - New Record

New menu name: Book

Remote Table: ☐

Columns

Table Columns for text

Column label	Type	Reference	Max length	Default value	Display
Title	String				false
Status	Choice				false
Author	String				false

Insert a new row...

Submit Cancel

Related Links

Step 3: Status Updated

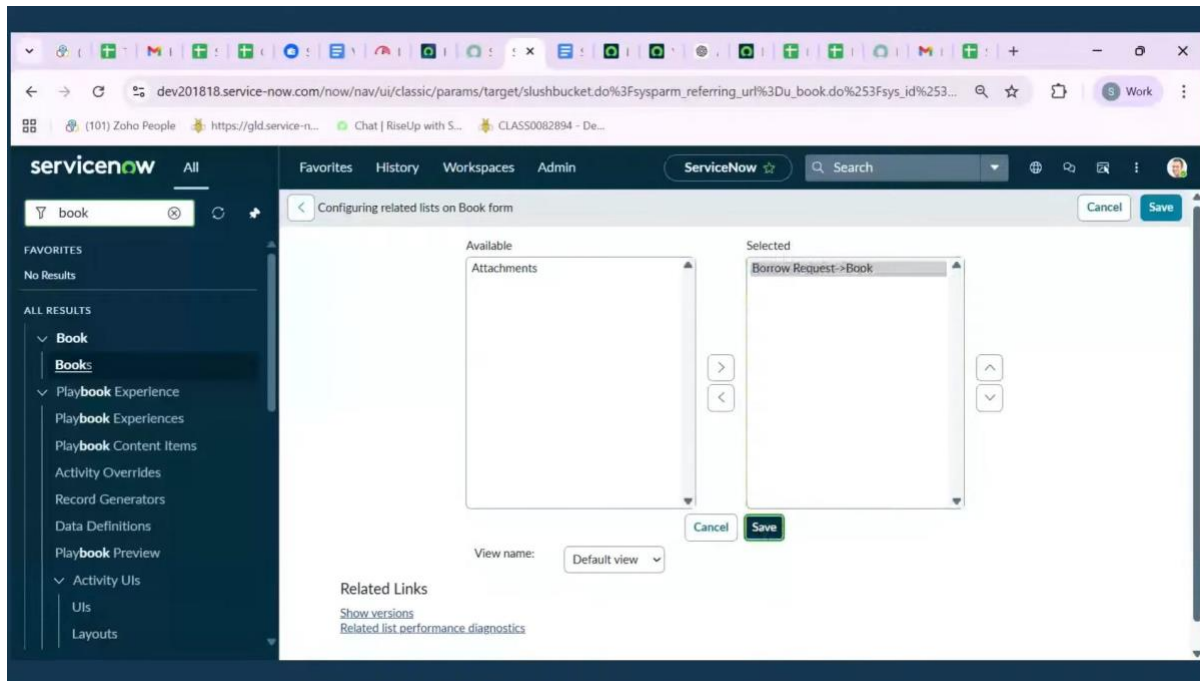
The screenshot shows the ServiceNow interface for configuring a Table Transform Map. The left sidebar contains a navigation menu with categories like 'System Import Sets', 'Administration', and 'Advanced'. The main panel is titled 'Table Transform Map - transform employee data'. It includes fields for 'Name' (transform employee data), 'Source table' (Import employee [u_import_e...]), 'Application' (Global), 'Created' (2024-10-28 21:54:35), 'Target table' (User [sys_user]), 'Order' (100), and 'Run script' (unchecked). There are also checkboxes for 'Active', 'Run business rules', 'Enforce mandatory fields', 'Copy empty fields', and 'Create new record on empty coalesce fields'. Below these are 'Copy', 'Update', and 'Delete' buttons. A 'Related Links' section lists 'Auto Map Matching Fields', 'Missing Asset', 'Validate Coalesce Fields', and 'Run Point Scan'. At the bottom, there is a table with columns 'Source field', 'Target field', and 'Coalesce', and a 'New' button.

The screenshot shows the ServiceNow interface for configuring a Table. The left sidebar contains a navigation menu with categories like 'Log Protection', 'System Archiving', 'System Clone', 'System Definition', and 'Tables'. The main panel is titled 'Table - Book'. It includes a table with columns 'Column label', 'Type', 'Reference', 'Max length', 'Default value', and 'Display'. The table contains the following data:

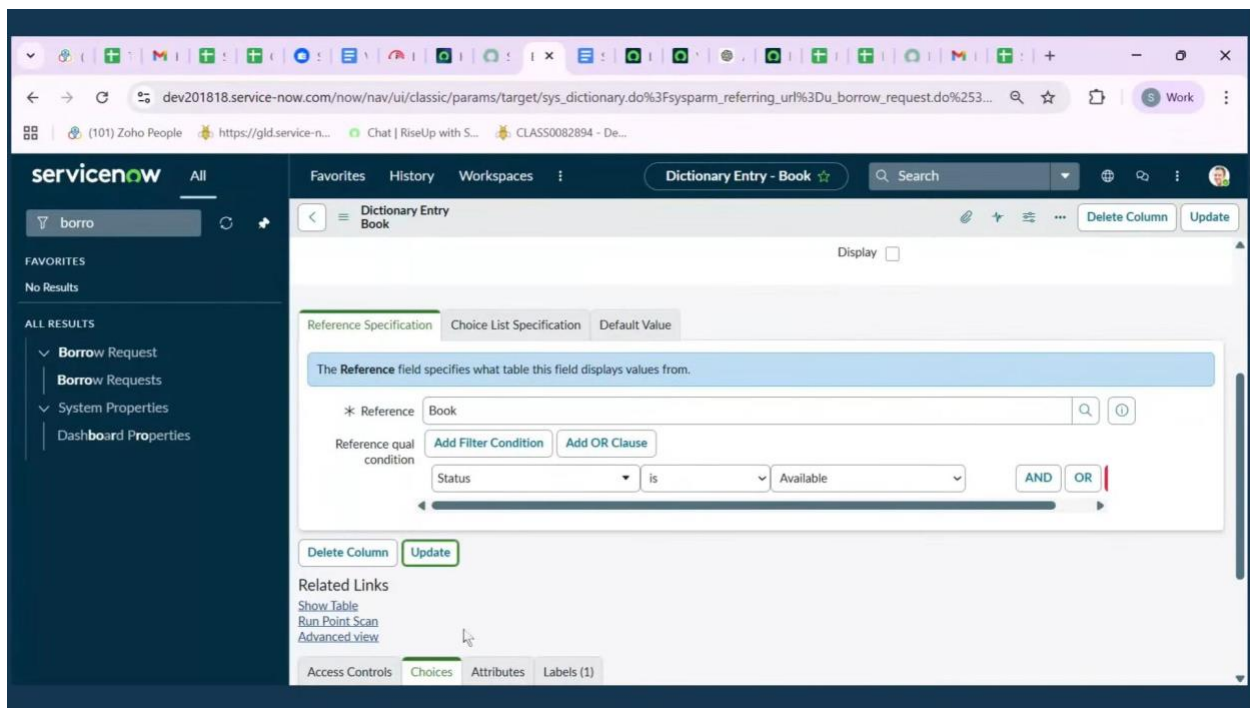
Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)	32		false
Updates	Integer	(empty)	40		false
Created	Date/Time	(empty)	40		false
Title	String	(empty)	40		false
Updated by	String	(empty)	40		false
Status	Choice	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Author	String	(empty)	40		false
Created by	String	(empty)	40		false

Below the table are 'Delete', 'Update', and 'Delete All Records' buttons. A 'Related Links' section lists 'Form Builder', 'Design Form', and 'Layout Form'.

Step 4: Configure Reference Qualifier



Step 5: Update Dictionary Book



Step 6: Flow Designer

The screenshot shows the 'New Flow' configuration page in ServiceNow Workflow Studio. The browser address bar displays the URL: `dev201818.service-now.com/now/workflow-studio/builder%3FtypeSysId%3D2d85e527439231106c4bb0117fb8f208%26sysId%3D-1%26`. The page title is 'Workflow Studio' and the tab is 'New Flow'. The main heading is 'Let's get the details for your flow'. The form includes:

- Flow name ***: A text input field containing 'Borrow request approval Flow'.
- Application ***: A dropdown menu set to 'Global'.
- Description**: A text area with the placeholder 'Describe your flow.'.
- Show additional properties**: A link to expand the configuration options.
- Buttons**: 'Cancel' and 'Build flow' buttons at the bottom right.

On the left, a canvas shows a visual representation of the flow with two steps, each represented by a blue bar and a plus icon.

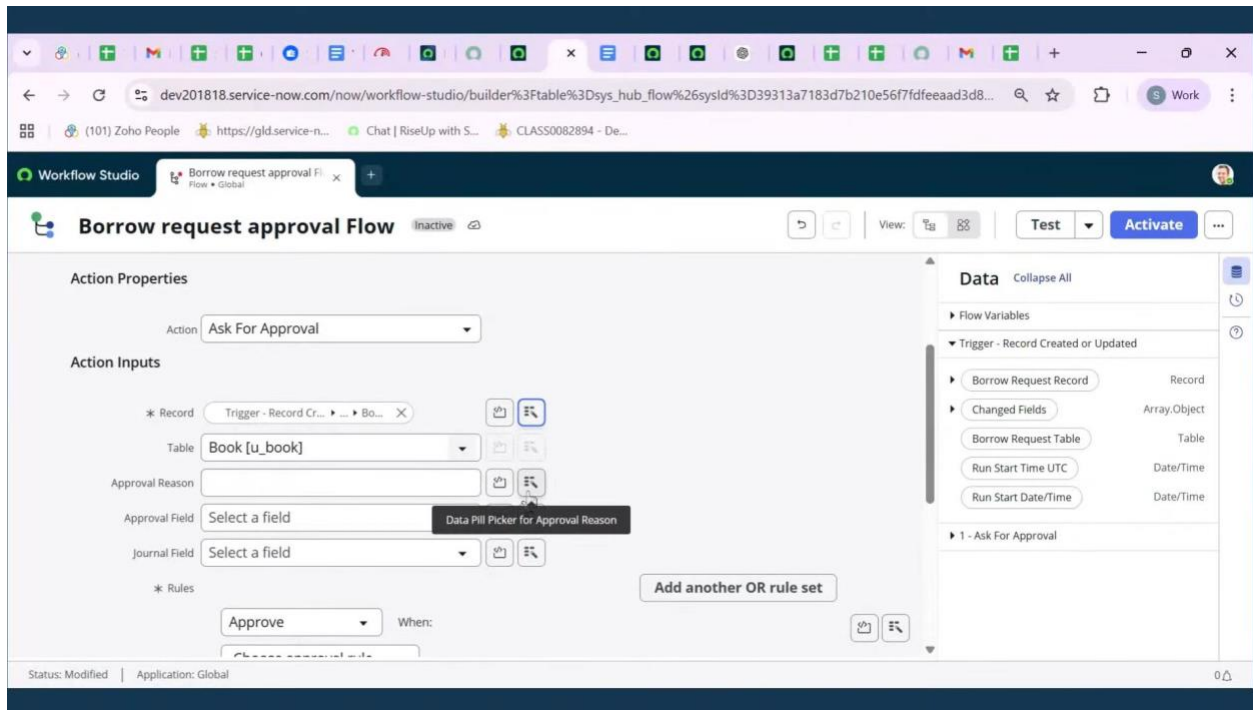
Step 7: Add a Trigger

The screenshot shows the 'Borrow request approval Flow' configuration page in ServiceNow Workflow Studio. The browser address bar displays the URL: `dev201818.service-now.com/now/workflow-studio/builder%3Ftable%3Dsys_hub_flow%26sysId%3D39313a7183d7b210e56f7dfeaad3d8...`. The page title is 'Workflow Studio' and the tab is 'Borrow request approval Flow'. The main heading is 'Borrow request approval Flow'. The flow is currently 'Inactive' and 'Saving...'. The 'Processing' button is disabled. The 'View' dropdown is set to 'Canvas'. The 'Test' button is disabled. The 'Activate' button is enabled. The 'Data' panel on the right shows the flow variables:

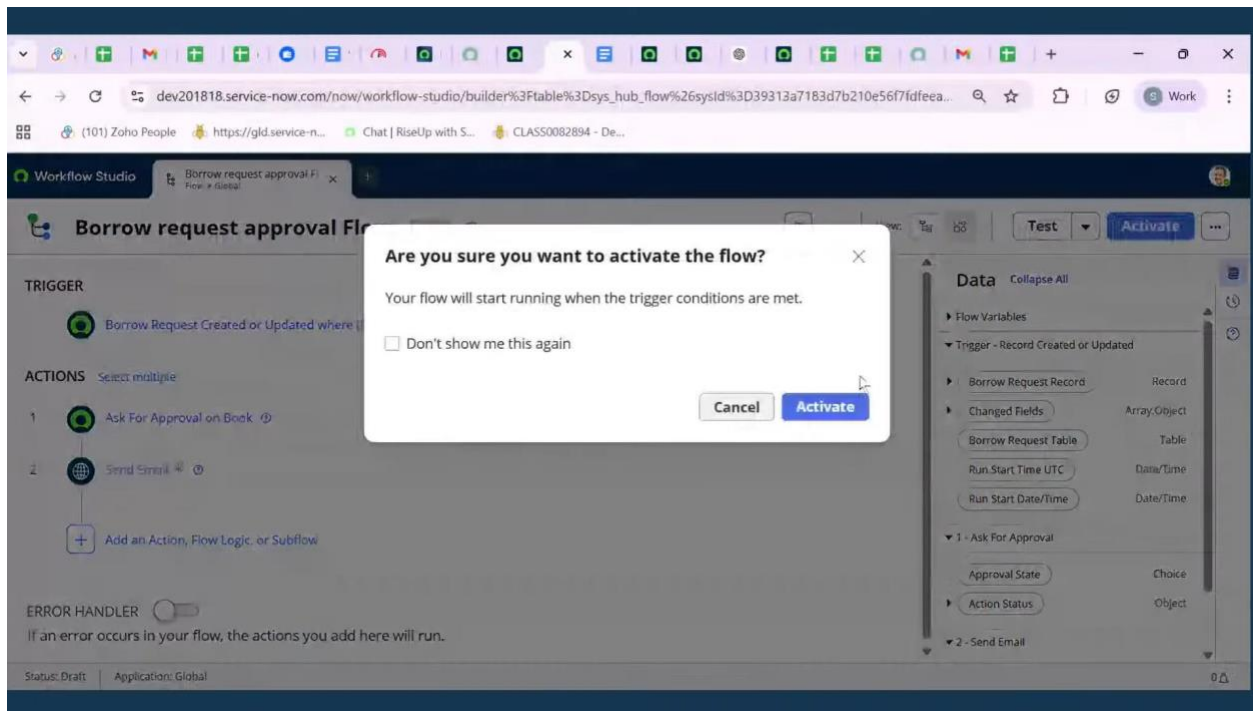
Flow Variables	Type
Trigger - Record Created or Updated	
Borrow Request Record	Record
Changed Fields	Array.Object
Borrow Request Table	Table
Run Start Time UTC	Date/Time
Run Start Date/Time	Date/Time

The 'TRIGGER' section shows a green circle icon and the text 'Borrow Request Created or Updated where (Status is Requested)'. The 'ACTIONS' section is empty, with a placeholder 'Add an Action, Flow Logic, or Subflow'. The 'ERROR HANDLER' section is also empty, with a placeholder 'If an error occurs in your flow, the actions you add here will run.'.

Step 8: Borrow Request Record



Step 9: Activate the Action



Step 10: UI Policy

The screenshot shows the ServiceNow 'UI Policy - New Record' page. The left sidebar contains a navigation menu with 'UI Policies' selected. The main content area has a header 'UI Policy - New Record' and a 'Submit' button. Below the header, there is a blue informational box explaining UI policies. The configuration section includes a 'Table' dropdown set to '-- None --', an 'Application' dropdown set to 'Global', and an 'Active' checkbox checked. A 'Short description' field contains the text 'Make Return Date mandatory when Issued.' and an 'Order' field set to '100'. At the bottom, there are tabs for 'When to Apply' and 'Script', with 'When to Apply' currently selected. Below the tabs are buttons for 'Add Filter Condition' and 'Add OR Clause', and a note about the policy's application scope.

UI policies change fields on a form based on a set of conditions. Use UI policies to show or hide fields, or to make fields mandatory or read-only based on these conditions. A UI policy specifies one or more actions to take when the policy is triggered. First, create the policy. Then add as many actions as necessary. UI policy actions are applied only if all of the following conditions are met:

- The UI Policy is **Active**
- The items in the **Conditions** field evaluate to true
- The field specified in the UI policy action is present on the specified form

[More info](#)

Table: -- None -- Application: Global

Active: ☒

* Short description: Make Return Date mandatory when Issued.

Order: 100

When to Apply: Script

Conditions: [Add Filter Condition](#) [Add OR Clause](#)

If selected, the UI Policy applies to all form views; otherwise the UI Policy is view-specific.

Apply the UI policy actions when the form is loaded and when the user changes values on the form.

Step 11: Create Report

The screenshot shows the ServiceNow 'Create a report' page. The left sidebar contains a navigation menu with 'reports' selected. The main content area has a header 'Create a report' and buttons for 'Save' and 'Run'. Below the header, there are tabs for 'Data', 'Type', 'Configure', and 'Style', with 'Data' currently selected. The 'Report name' field contains 'Most Borrowed Books.' and the 'Table' dropdown is set to 'Borrow Request [u_borrow_request]'. A 'Description' field contains the text 'There is no description for this table. To add a description, please contact your admin.' At the bottom, there is a 'Next' button. On the right side, there is a section titled 'Table: Borrow Request [u_borrow_request]' with a 'All' filter and a 'Create your report with Analytics Q&A' section. This section includes a text input field 'What do you want to see?' and an 'Ask' button.

Create a report

Report Title: Most Borrowed Books.

* Report name: Most Borrowed Books.

* Source type: Table

* Table: Borrow Request [u_borrow_request]

Description: There is no description for this table. To add a description, please contact your admin.

Next

Table: Borrow Request [u_borrow_request]

All

Create your report with Analytics Q&A

Ask for information. You can give simple filtering conditions. You get the answer with an appropriate visualization.

What do you want to see? Ask

How can I improve my results?

dev201818.service-now.com/now/nav/ui/classic/params/target/sys_report_template.do%3Fjvar_report_id%3D2e193a31831bb210e56f7fdf...

Work

servicenow

All

reports

FAVORITES

No Results

ALL RESULTS

Platform Analytics Administrat...

Data Sources

Report Sources

Usage and governance

Reports

Report Statistics

Schedules

Scheduled Reports

Color Settings

Report Ranges

Properties

Reporting Properties

Favorites

History

Workspaces

Admin

Most Borrowed Books

Search

Edit report

Save

Run

Data

Type

Configure

Style

Report Title :

Most Borrowed Books

Group by

Book

Additional group by

Stack by

-- None --

Display data table

Configuration field

Aggregation

Count

Set Value Formatting

Max number of groups

Back

Next

Type a question about your data

What do you want to see?

Ask

How can I improve my results?

To modify the current report, use the left panel or Edit Condition.

Table: Borrow Request [u_borrow_request]

All

No data to display

8. PROJECT DEMONSTRATION:

The project demonstration phase is where the completed system is presented and explained in Video Format

- Show the Book table and available books.
- Login as a Student and submit a borrow request.
- Show the request status as Requested.
- Login as a Librarian and approve the request.
- Show that the book status changes to Issued.
- Display the Most Borrowed Books report

FINAL RESULT:

- The system successfully automates the library book request process, improves efficiency, and provides better tracking of library resources

9. CONCLUSION:

- The Smart Library Request Workflow in ServiceNow simplifies the library management process by automating book requests, approvals, and tracking.
- This improves efficiency for both students and librarians.